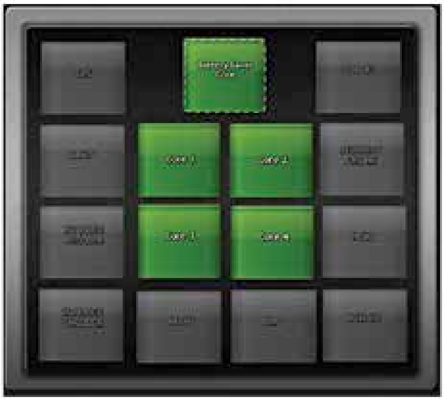
rendering, watching HD movies, gaming and so on. Its current iteration – the Tegra 4 – includes a quad-core ARM Cortex A15 plus one companion core.

Apple makes its own custom SoCs for the iOS products. The Apple A4 (housing a single core Cortex A8 processor) is the oldest SoC still seen in current 4th gen iPod whereas the most recent iteration is the A6 which is seen in the iPhone 5 and A6X seen in the iPad 4th gen. Apple A5 (housing a dual-core Cortex A9) is the most popular SoC as it's found on quite a lot of Apple devices currently in the market. Apple is probably the only SoC maker who hasn't yet come out with a quad-core processor configuration. Since it has complete control over both the software and hardware on its devices and since Apple SoCs are seen in its limited number of iDevices, we are seeing more power efficient SoCs, better graphics on their SoCs to handle the high-res Retina displays and high graphics iOS games and so on.



Samsung has a limited number of SoCs from its Exynos lineup which are seen mostly on their devices. It first came out with the Exynos 4 Quad with the quad-core Cortex A9 was seen in the Galaxy Note 10.1, Galaxy Note 8.0 and Galaxy Note II. The current generation includes the Exynos 5 Dual (dual-core Cortex A15 configuration) and the Exynos 5 Octa (quad-core Cortex A15 + quad-core Cortex A7 in the bit.LITTLE configuration). The Exynos 5 Octa is seen in Samsung Galaxy S4 and its processor cores work in the big.LITTLE configuration – lower power Cortex A7 takes over when running in the low power mode, whereas the higher end Cortex A15 processors are employed for heavy tasks such as gaming. So even though the SoC has the Octa naming convention, at a time you will have four cores working.

Off late we are seeing a lot of affordable smartphones which come with dual-core and sometimes even quad-core configurations. One look at the specs sheet of such devices and you will find one commonality - MediaTek. Its flagship model - the MT6589 SoC is seen in Micromax Canvas HD A116 which had us impressed for its value for money proposition.